

annotators are more likely to predict that a conversation will end when it actually does end.

Multilingual Prompts and Models The choice of model and prompting strategy has a strong effect on the F1 score across languages, as shown in Figure 10b. As discussed in §5, these results are crucial for understanding when LLMs are effective SIMULATORS.

As in §5, the differences are largely driven by how often the model or prompt predicts conversation endings. Similarly, the COT prompt predicts conversation endings most frequently (13% for English, 6% for Chinese, 16% for Russian), while the OVERRIDE prompt predicts them least frequently (1% for English, 0.5% for Chinese, and 3% for Russian). In this analysis, we compare three prompts, one from each category. In all languages, the OVERRIDE prompt results in lower F1 scores, while the COT prompt yields higher F1 scores. Although the results from one prompt of each variety cannot be generalized to all COT and OVERRIDE prompts, future work may examine whether these differences are attributable to the structure of these prompts. For instance, the COT prompt explicitly asks the model to reason whether HUMAN achieved the goal satisfactorily, which may lead the model to prioritize the decision to end the conversation. In contrast, the OVERRIDE prompt, which tricks the model into performing the task, does not explicitly prompt the model to end the conversation as often.

Interestingly, these results contrast with the performance of prompts in matching the properties of the response text. In many metrics, the COT performs worse than the DIRECT or OVERRIDE prompts. Again, future research may explore whether these differences generalize to a broader class of COT prompts. Perhaps the more structured format of the COT prompt may be less suited to capturing the nuances of open-ended human speech, even though it might be better for closed-ended tasks.

The correlations for each individual metric, by prompt and model, are given in Table 7 and Table 8.

Simulate a person using an LLM. Respond like a regular person. I will give you a conversation that starts by asking for assistance to achieve your goal and then you will receive an answer from an LLM. If you goal achieved, you should not respond. If not, you will write the person's next response based on both previous turns! Generate only the last human response.

You said this to an LLM:

Generate an etsy title for a art work that can be added to mugs, painting, tshirt, etc within 140 characters with no & only | to separate without first word being whimsical and is the name of the artMonarch

LLM response:

"Monarch Majesty | Butterfly Artwork for Elegant Mugs, Stylish T-Shirts, and Refined Paintings | Versatile Home Decor & Fashion"

Will you respond?	If yes, write the person's next response.
☒ [1] Yes ☐ [2] No ☐ [3] Non-English Content (can't answer) ☐ [4] Uncomfortable Answering (NSFW, persona you don't want to adopt, etc.)	Can you generate some a more creative title that catches attention?

Move backward

Move forward

Figure 8: Annotation interface for annotators to infer human Turn 3

Category	Measure	Description
End	F1	Comparison of how often the SIMULATOR ends the conversation whens the human ends it.
Lexical	Utterance Length ^s Average Word Length ^s Perplexity ^s Typographical Errors ^s	Log-transformed number of words. Log-transformed number of characters per word. Log-transformed perplexity of the utterance, calculated using lmpp1 and GPT-2 as model. Fraction of words that have typographical errors, counted using pyspellchecker.
Syntactic	Part of Speech ^d Dependency Tree Depth ^s Tree Breadth ^s Tree Dependency Distance ^s	Distribution of the utterance’s part of speech tags from spaCy. Log-transformed depth of the dependency tree from spaCy. Log-transformed number of leaf nodes. Log-transformed average distance between dependents.
Semantic	SBERT ^v LIWC ^d Prompt Type ^d	Cosine similarity of utterance embeddings from the all-MiniLM-L6-v2 language model (Reimers and Gurevych, 2019). Distribution of 69 LIWC categories from Pennebaker et al. (2007). Distribution of categories from prompt classification tool valpy/prompt-classification.
Style	Punctuation ^d Capitalization ^s Sentiment ^s Politeness ^s Formality ^s Toxicity ^s Readability ^s Subjectivity ^s LUAR ^v	Distribution of punctuation characters. Fraction of letters that are capitalized. Distribution of positive, neutral, and negative sentiment from distilbert-base-multilingual-cased-sentiments-student. From Genius1237/xlm-roberta-large-tydip (Srinivasan and Choi, 2022). From s-nlp/mdeberta-base-formality-ranker (Dementieva et al., 2023). Toxicity of tone and content, as judged by annotators s-nlp/roberta_toxicity_classifier. Distribution of Flesch reading ease scores (Flesch, 1979). The average subjectivity score of words in the utterance from the sentiment polarity lexicon in textblob. Author style embeddings using rriviera1849/LUAR-CRUD (Soto et al., 2021).

Table 1: Measures used to evaluate how well LLMs capture properties of human responses at Turn 3 of a conversation. Letter superscript indicates whether the difference between human and SIMULATOR measurements are (s) scalar values (compared with l1-distance), (d) probability distributions (compared with Jensen-Shannon divergence), or (v) vector embeddings (compared with cosine distance).